

# The Very Basic Note in Combinatorics

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# 1 Set Notations

**Definition 1.1.** A *set*(集合) is a collection of certain objects. The things belonging to a set are called the *elements*(元素/元) of the set.

In most cases, the elements of a set are simply numbers. We use  $S = \{a, b, c, \dots\}$  to represent a set  $S$ , where  $a, b, c, \dots$  are the elements of  $S$ . We write  $a \in S$  to mean  $a$  belongs to  $S$ . If  $x$  does not belong to  $S$ , we write  $x \notin S$ .

Another common way to represent a set is to write  $S = \{a : \dots\}$ . This means  $S$  is the collection of all elements  $a$  that satisfy certain properties as indicated in the  $\dots$  part.

**Definition 1.2.** Let  $A$  and  $B$  be any sets. We say that  $A$  and  $B$  are the same, and write  $A = B$ , if each element in  $A$  belongs to  $B$ , and each element in  $B$  belongs to  $A$ .

In other words, two sets are equal if they contain the same elements. Here, repeated elements are regarded as the same, and the order does not matter. For example, the sets  $\{1, 1, 2, 2, 3\}$ ,  $\{2, 3, 1, 3\}$  and  $\{1, 2, 3\}$  are the same.

The following are the notations of some commonly used sets.

- $\mathbb{R}$ : the set of all real numbers
- $\mathbb{Q}$ : the set of all rational numbers
- $\mathbb{Z}$ : the set of all integers
- $\mathbb{N}$ : the set of all positive integers
- $\mathbb{R}^+$ : the set of all positive real numbers
- $\mathbb{Q}^+$ : the set of all positive rational numbers
- $\mathbb{R}_0^+$ : the set of all nonnegative real numbers
- $\mathbb{N}_0$ : the set of all nonnegative integers
- $\emptyset$ : the *empty set*(空集), which is the set containing no element

Note that the notations  $\mathbb{R}_0^+$  and  $\mathbb{N}_0$  are not that commonly used.

**Example 1.1.** The set  $\{n \in \mathbb{Z} : 3 \mid n\}$  is the collection of all integers which are divisible by 3, while  $\{n \in \mathbb{N} : n = m^2 \text{ for some } m \in \mathbb{N}\}$  is the set of all positive square numbers.

**Definition 1.3.** A set is an *infinite set*(無限集/無窮集) if it contains infinitely many elements, and is a *finite set*(有限集/有窮集) if it contains finitely many elements. In the latter case, the number of elements of the set  $S$  is called the *order*(濃度) or *cardinality* of the set and is denoted by  $|S|$ .

In the following, when we write  $|S|$  for a set  $S$ , we implicitly mean that  $S$  is a finite set.

**Example 1.2.** All of  $\mathbb{R}, \mathbb{Q}, \mathbb{Z}, \mathbb{N}$  are infinite sets. The set  $S = \{1, 2, 4\}$  has order 3, so that we write  $|S| = 3$ . Also,  $|\emptyset| = 0$ .

**Definition 1.4.** Let  $A$  and  $B$  be any sets. If every element in  $A$  belongs to  $B$ , then  $A$  is called a *subset*(子集) of  $B$ , and we use the notation  $A \subset B$  or  $A \subseteq B$ . In addition, if  $A \neq B$ , then  $A$  is called a *proper subset*(真子集) of  $B$ , and we use the notation  $A \subsetneq B$ .

**Example 1.3.** We have a chain of subsets  $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$  by definition. The set  $\{1, 2, 4\}$  is a proper subset of  $\{1, 2, 3, 4, 5\}$ . Note that the empty set is a subset of every other set while every set is a subset of itself.

**Proposition 1.1.** Let  $A$  and  $B$  be any sets. Then  $A = B$  if and only if  $A \subset B$  and  $B \subset A$ .

This follows easily from the definitions of equality of sets and subset relations.

**Proposition 1.2.** Let  $A, B, C$  be any sets. If  $A \subset B$  and  $B \subset C$ , then  $A \subset C$ .

*Proof.* For any element  $a \in A$ , since  $A \subset B$ , we have  $a \in B$ . Similarly, this implies  $a \in C$ . As this holds for every  $a \in A$ , we get  $A \subset C$  by definition.  $\square$

**Definition 1.5.** Let  $A$  and  $B$  be any sets. Their *union*(並集)  $A \cup B$  is defined to be the set containing all elements belonging to at least one of  $A$  and  $B$ .

**Definition 1.6.** Let  $A$  and  $B$  be any sets. Their *intersection*(交集)  $A \cap B$  is defined to be the set containing all elements belonging to both  $A$  and  $B$ .

In general, we write  $S_1 \cup S_2 \cup \dots \cup S_n$  to denote the set containing all elements belonging to at least one of  $S_1, S_2, \dots, S_n$ . Similarly, we write  $S_1 \cap S_2 \cap \dots \cap S_n$  to denote the set containing all elements belonging to all of  $S_1, S_2, \dots, S_n$ . Two sets are called *disjoint*(互斥) if their intersection is the empty set.

**Example 1.4.** Let  $A = \{1, 2, 4\}$ ,  $B = \{1, 3, 4, 5\}$  and  $C = \{4, 5, 6\}$ . Then we have

$$\begin{aligned} A \cap B &= \{1, 4\}, \\ A \cup B &= \{1, 2, 3, 4, 5\}, \\ A \cap B \cap C &= \{4\}, \\ A \cup B \cup C &= \{1, 2, 3, 4, 5, 6\}. \end{aligned}$$

The sets  $A$  and  $\{3, 5, 6\}$  are disjoint.

**Proposition 1.3.** Let  $A, B, C$  be any sets. Then the following hold.

- (a)  $A \cup A = A$
- (b)  $A \cup \emptyset = A$
- (c)  $A \subset (A \cup B)$
- (d)  $A \cap A = A$
- (e)  $A \cap \emptyset = \emptyset$
- (f)  $(A \cap B) \subset A$
- (g)  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- (h)  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- (i)  $|A \cup B| = |A| + |B| - |A \cap B|$

*Proof.* Most results are immediate consequences of the definitions. We only provide a proof of (g) as an example.

For any  $a \in (A \cap (B \cup C))$ , we have  $a \in A$  and  $a \in (B \cup C)$ . Without loss of generality, assume  $a \in B$ . Then  $a \in (A \cap B)$ , and hence  $a \in (A \cap B) \cup (A \cap C)$ . This proves  $(A \cap (B \cup C)) \subset ((A \cap B) \cup (A \cap C))$ .

For any  $a \in ((A \cap B) \cup (A \cap C))$ , without loss of generality, assume  $a \in (A \cap B)$ . This implies  $a \in A$  and  $a \in B$ . Thus,  $a \in (B \cup C)$ , and hence  $a \in (A \cap (B \cup C))$ . This proves  $((A \cap B) \cup (A \cap C)) \subset (A \cap (B \cup C))$ .

By [proposition 1.1](#), we get  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ . □

**Definition 1.7.** Let  $A$  and  $B$  be any sets. The *complement*(差集) of  $A$  in  $B$ , denoted by  $B - A$  or  $B \setminus A$ , is the set containing all elements belonging to  $B$  but not to  $A$ .

**Example 1.5.** Let  $A = \{1, 2, 4\}$  and  $B = \{1, 3, 4, 5\}$ . Then we have  $A - B = \{2\}$  and  $B - A = \{3, 5\}$ .

**Proposition 1.4.** Let  $A, B, C$  be any sets. Then the following hold.

- (a)  $A - A = \emptyset$
- (b)  $A - \emptyset = A$
- (c)  $\emptyset - A = \emptyset$
- (d)  $A - B = \emptyset$  if and only if  $A \subset B$
- (e)  $(A - B) \cap B = \emptyset$
- (f)  $A = (A - B) \cup (A \cap B)$
- (g)  $A - (B \cup C) = (A - B) \cap (A - C)$
- (h)  $A - (B \cap C) = (A - B) \cup (A - C)$
- (i)  $|A - B| = |A| - |A \cap B|$ .

*Proof.* Most results are immediate consequences of the definitions. We only provide a proof of (g) as an example. Indeed, we have

$$\begin{aligned}
 & a \in (A - (B \cup C)) \\
 \Leftrightarrow & a \in A \text{ and } a \notin (B \cup C) \\
 \Leftrightarrow & a \in A \text{ and } a \notin B \text{ and } a \notin C \\
 \Leftrightarrow & a \in (A - B) \text{ and } a \in (A - C) \\
 \Leftrightarrow & a \in ((A - B) \cap (A - C)).
 \end{aligned}$$

This proves  $A - (B \cup C) = (A - B) \cap (A - C)$ . □

**Definition 1.8.** Let  $A$  and  $B$  be any sets. Their *symmetric difference*(相對差集)  $A \Delta B$  is the set containing all elements belonging to exactly one of  $A$  and  $B$ .

**Example 1.6.** Let  $A = \{1, 2, 4\}$ ,  $B = \{1, 3, 4, 5\}$  and  $C = \{4, 5, 6\}$ . Then we have

$$A \Delta B = \{2, 3, 5\} \quad \text{and} \quad A \Delta C = \{1, 2, 5, 6\}.$$

**Proposition 1.5.** Let  $A$  and  $B$  be any sets. Then the following hold.

- (a)  $A \Delta A = \emptyset$
- (b)  $A \Delta \emptyset = A$
- (c)  $A \Delta B = (A \cup B) - (A \cap B)$
- (d)  $|A \Delta B| = |A| + |B| - 2|A \cap B|$

*Proof.* Most results are immediate consequences of the definitions. We only provide a proof of (d) as an example. We need to count the number of elements in  $A \Delta B$ . The total number of elements in  $A$  and  $B$  is  $|A| + |B|$ . However, those elements in both  $A$  and  $B$ , i.e. those elements in  $A \cap B$ , should not be counted. Each such element has been counted twice. This proves  $|A \Delta B| = |A| + |B| - 2|A \cap B|$ .  $\square$

**Definition 1.9.** Let  $A$  and  $B$  be any sets. The *Cartesian product*(直積)  $A \times B$  is the set containing all ordered pairs  $(a, b)$  where  $a \in A$  and  $b \in B$ .

Similarly, we can define the Cartesian product  $S_1 \times S_2 \times \cdots \times S_n$  of more sets.

**Example 1.7.** Let  $A = \{1, 2, 4\}$  and  $B = \{1, 5\}$ . Then we have

$$A \times B = \{(1, 1), (1, 5), (2, 1), (2, 5), (4, 1), (4, 5)\}.$$

**Proposition 1.6.** Let  $A$  and  $B$  be any finite sets. Then  $|A \times B| = |A||B|$ .

*Proof.* There are  $|A|$  ways to choose the first entry of the ordered pair  $(a, b)$ , and  $|B|$  ways to choose the second entry. Therefore, there are  $|A||B|$  ordered pairs. This implies  $|A \times B| = |A||B|$ .  $\square$

**Definition 1.10.** The *universal quantifier*(全稱量詞)  $\forall$  means ‘for all’, while the *existential quantifier*(存在量詞)  $\exists$  means ‘there exist’.

**Example 1.8.** The statement  $\forall x \in \mathbb{R}, x^2 \geq 0$  means ‘for all real numbers  $x$ ,  $x^2 \geq 0$ ’. The statement  $\exists n \in \mathbb{Z}, 2 \mid n$  means ‘there exists an integer  $n$  which is divisible by 2’.

## 2 Counting and the Pigeonhole Principle

**Definition 2.1.** The *method of exhaustion*(窮舉法/枚舉法) refers to the counting method which lists out all possible cases.

**Theorem 2.1.** (*Addition principle/Sum rule*(加法原理)) Let  $S_1, S_2, \dots, S_n$  be pairwise disjoint sets. Then we have

$$|S_1 \cup S_2 \cup \dots \cup S_n| = |S_1| + |S_2| + \dots + |S_n|.$$

*Proof.* This is obvious since every element in  $S_1 \cup S_2 \cup \dots \cup S_n$  belongs to exactly one of  $S_1, S_2, \dots, S_n$ .  $\square$

For the application in counting, it simply means that we can count by partitioning all possibilities into a few cases. Each case has its own feature which allows us to count quickly by using some other methods. Note that we should ensure that each situation falls into exactly one case in order to apply the addition principle (which is the disjoint condition).

**Example 2.1.** There are 12 girls and 10 boys in a class. Then there are  $12 + 10 = 22$  students in total. This is the simplest application of the addition principle.

**Theorem 2.2.** (*Multiplication principle/Product rule*(乘法原理)) Let  $S_1, S_2, \dots, S_n$  be any sets. Then we have

$$|S_1 \times S_2 \times \dots \times S_n| = |S_1| |S_2| \dots |S_n|.$$

This is a generalization of [proposition 1.6](#).

For the application in counting, suppose we would like to make some choices in a few steps so that the number of choices in each step is independent of the other steps. Then we can simply take the product of the numbers of choices in all steps to obtain the total.

**Example 2.2.** There are 12 girls and 10 boys in a class. Then there are  $12 \times 10 = 120$  ways to choose a girl and a boy as the monitors. This is the simplest application of the multiplication principle.

**Definition 2.2.** (*Permutation*(排列)) Let  $n$  and  $r$  be nonnegative integers satisfying  $0 \leq r \leq n$ . Then the number of ways to choose  $r$  objects from  $n$  distinct objects in order is denoted by  $P_r^n$ .



**Definition 2.3.** (*Combination*(組合)) Let  $n$  and  $r$  be nonnegative integers satisfying  $0 \leq r \leq n$ . Then the number of ways to choose  $r$  objects from  $n$  distinct objects without specifying the order is denoted by  $\binom{n}{r}$  or  $C_r^n$ .

In CMO and CGMO, it is common to use  $P_n^r$  (or  $A_n^r$ ) to denote the number of permutations and  $C_n^r$  to denote the number of combinations (with  $n$  and  $r$  being swapped). The symbol  $\binom{n}{r}$  is also known as the *binomial function*(二項式函數) or the *binomial coefficient*(二項式係數), due to the following theorem.

**Theorem 2.3.** (*Binomial theorem*(二項式定理)) Let  $n$  be a nonnegative integer. Then

$$(x + y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 + \cdots + \binom{n}{n}y^n.$$

*Proof.* See *theorem 1.1* of **Techniques in Combinatorics 3.1.1**. □

**Proposition 2.1.** Let  $n$  and  $r$  be nonnegative integers satisfying  $0 \leq r \leq n$ . Then the following hold.

- (a)  $P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$
- (b)  $P_n^n = n!$
- (c)  $P_1^n = n$
- (d)  $P_0^n = 1$

*Proof.* See *proposition 1.3* of **Techniques in Combinatorics 3.1.1**. □

Note that  $0!$  is defined to be 1.

**Example 2.3.** There are 22 students in a class. Then there are

$$P_5^{22} = 22 \times 21 \times 20 \times 19 \times 18 = 3160080$$

ways to choose 5 different students to be on duty from Monday to Friday (with one student on duty on each day).

**Proposition 2.2.** Let  $n, r, k$  be nonnegative integers satisfying  $0 \leq r \leq n$ . Then the following hold.

$$(a) \binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{r!(n-r)!}$$

$$(b) \binom{n}{r} = \binom{n}{n-r}$$

$$(c) \binom{n}{n} = \binom{n}{0} = 1$$

$$(d) \binom{n}{1} = n$$

$$(e) \text{ (Pascal's identity (帕斯卡恒等式)) } \binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1} \text{ for } r \geq 1$$

$$(f) \binom{n}{n} + \binom{n+1}{n} + \cdots + \binom{n+k}{n} = \binom{n+k+1}{n+1}$$

$$(g) \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

*Proof.* See [proposition 1.4](#) of **Techniques in Combinatorics 3.1.1**. □

**Example 2.4.** There are 22 students in a class. Then there are

$$\binom{22}{3} = \frac{22 \times 21 \times 20}{3 \times 2 \times 1} = 1540$$

ways to choose 3 students to work on a project together.

**Theorem 2.4.** (Inclusion-exclusion principle (容斥原理)) Let  $S_1, S_2, \dots, S_n$  be any finite sets. Then

$$\begin{aligned} & |S_1 \cup S_2 \cup \cdots \cup S_n| \\ &= \sum_{i=1}^n |S_i| - \sum_{i < j} |S_i \cap S_j| + \sum_{i < j < k} |S_i \cap S_j \cap S_k| - \cdots + (-1)^{n+1} |S_1 \cap S_2 \cap \cdots \cap S_n|. \end{aligned}$$

*Proof.* See [theorem 1.2](#) of **Techniques in Combinatorics 3.1.1**. □

In particular, we have

$$|A \cup B| = |A| + |B| - |A \cap B|,$$

which has been given in [proposition 1.3\(i\)](#), and

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

**Example 2.5.** In a class, 6 students get 5\*\* in the exam of Mathematics Core, 3 students get 5\*\* in the exam of M2, and 1 student get 5\*\* in both exams. By the inclusion-exclusion principle, exactly  $6 + 3 - 1 = 8$  students get 5\*\* in at least one of these exams.

**Theorem 2.5.** (Pigeonhole principle(抽屜原理/鴿巢原理)) Suppose there are  $n$  elements each of which belongs to exactly one of the sets  $S_1, S_2, \dots, S_m$ . Then there exist  $\left\lceil \frac{n}{m} \right\rceil$  of them which belong to the same set.

Here,  $\lceil x \rceil$  is the ceiling function, which denotes the smallest integer not less than  $x$ .

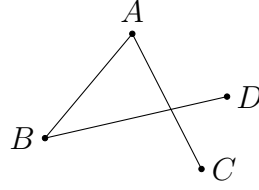
*Proof.* Suppose on the contrary that each  $S_j$  contains less than  $\left\lceil \frac{n}{m} \right\rceil$  elements. This implies each  $S_j$  contains at most  $\frac{n-1}{m}$  elements. Thus, the total number of elements is at most  $\frac{n-1}{m} \cdot m = n-1$ , which is a contradiction.  $\square$

**Example 2.6.** There are 25 students in a class. By the pigeonhole principle,  $\left\lceil \frac{25}{12} \right\rceil = 3$  of the students must be born in the same month.

### 3 Graph Theory

**Definition 3.1.** A *simple graph*(簡單圖)  $G = (V, E)$  consists of a nonempty set of *vertices*(頂點)  $V$  and a set of *edges*(邊)  $E$  containing some unordered pairs of distinct vertices, such that there is at most one edge between every pair of vertices.

**Example 3.1.** Let  $V = \{A, B, C, D\}$  and  $E = \{\{A, B\}, \{A, C\}, \{B, D\}\}$ . Then the simple graph  $G = (V, E)$  consists of 4 vertices  $A, B, C, D$  and 3 edges.



Note that  $\{A, B\}$  and  $\{B, A\}$  are the same edge. Also,  $\{A, A\} = \{A\}$  is not regarded as an edge in a simple graph.

We can visualize a simple graph geometrically. The vertices are represented by points, and edges are represented by segments joining the vertices. Since only the endpoints matter, we may use curves instead of segments.

**Definition 3.2.** Let  $G = (V, E)$  be a simple graph. The *degree*(度) of a vertex  $A \in V$  is the number of edges having  $A$  as one of the endpoints, and is denoted by  $\deg A$ .

**Example 3.2.** Let  $V = \{A, B, C, D\}$  and  $E = \{\{A, B\}, \{A, C\}, \{B, D\}\}$ . Then  $\deg A = \deg B = 2$  and  $\deg C = \deg D = 1$ .

**Proposition 3.1.** Let  $G = (V, E)$  be a simple graph. Then the following hold.

- (a)  $0 \leq \deg A \leq |V| - 1$  for any vertex  $A \in V$
- (b) (*Handshaking theorem*(握手定理))  $\sum_{A \in V} \deg A = 2|E|$

*Proof.*

- (a) The lower bound is trivial. The upper bound holds since there are  $|V| - 1$  vertices other than  $A$ .
- (b) For each edge in  $E$ , it contributes 2 to the sum of degrees as it has 2 endpoints. The result follows immediately.

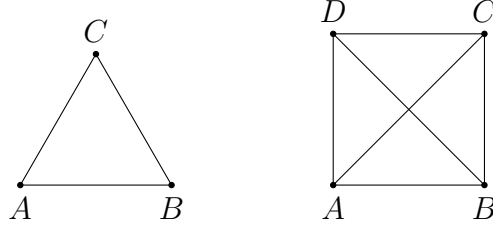
□

**Example 3.3.** Let  $V = \{A, B, C, D\}$  and  $E = \{\{A, B\}, \{A, C\}, \{B, D\}\}$ . Then we have

$$\deg A + \deg B + \deg C + \deg D = 2 + 2 + 1 + 1 = 6 = 2|E|.$$

This verifies the handshaking theorem in this case.

**Definition 3.3.** A *complete graph*(完全圖) of  $n$  vertices is a simple graph for which every pair of vertices is joined by an edge. It is denoted by  $K_n$ .



**Proposition 3.2.** In any complete graph  $K_n$ , there are  $\binom{n}{2}$  edges in total.

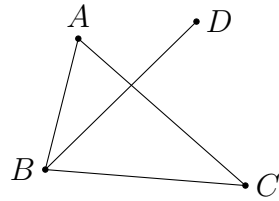
*Proof.* This is because there are  $\binom{n}{2}$  ways to choose 2 unordered vertices. □

**Definition 3.4.** A *walk*(路徑) of length  $s$  is a sequence of edges  $e_1, e_2, \dots, e_s$  in a simple graph such that the terminal vertex of  $e_j$  coincides with the initial vertex of  $e_{j+1}$  for each  $j = 1, 2, \dots, s-1$ .

**Definition 3.5.** A *circuit*(迴路) of length  $s$  is a walk  $e_1, e_2, \dots, e_s$  in a simple graph such that the terminal vertex of  $e_s$  also coincides with the initial vertex of  $e_1$ .

We sometimes use a sequence of vertices  $V_1, V_2, \dots, V_{s+1}$  to denote a walk of length  $s$ , where  $V_j$  and  $V_{j+1}$  are joined by an edge for each  $j = 1, 2, \dots, s$ . A *path*(簡單路徑) is a walk for which every vertex is travelled at most once (possibly except the initial and terminal vertex). A *connected graph*(連通圖) means a graph for which every two vertices are connected by a path. Also,  $V_1, V_2, \dots, V_s, V_1$  denotes a circuit of length  $s$ , and a *cycle*(圈) is a circuit for which every vertex is travelled at most once except the initial and terminal vertex.

**Example 3.4.** For the simple graph as shown,  $D, B, C, A$  is a path of length 4, while  $A, B, C, A$  is a cycle of length 3. Also,  $A, B, D, B, C, A, B$  is a walk.



# Links

## Theorems

- Addition principle
- Binomial theorem
- Handshaking theorem
- Inclusion-exclusion principle
- Multiplication principle
- Pascal's identity
- Pigeonhole principle
- Product rule
- Sum rule

## Propositions

- 1.1: Equality of sets
- 1.2: Transitivity of subset relations
- 1.3: Properties of union and intersection of sets
- 1.4: Properties of complement of sets
- 1.5: Properties of symmetric difference of sets
- 1.6: Order of Cartesian product of sets
- 2.1: Properties of permutations
- 2.2: Properties of combinations
- 3.1: Properties of degrees in simple graphs
- 3.2: Number of edges in complete graphs

## Terminologies and Notations

- $a \in S$
- $a \notin S$
- $A = B$  (set)
- $\mathbb{N}$
- $\mathbb{N}_0$
- $\mathbb{Q}$
- $\mathbb{Q}^+$

- $\mathbb{R}$
- $\mathbb{R}^+$
- $\mathbb{R}_0^+$
- $\mathbb{Z}$
- binomial coefficient  $\binom{n}{r}$
- binomial function  $\binom{n}{r}$
- cardinality  $|S|$
- Cartesian product  $A \times B$
- circuit
- combination  $\binom{n}{r}$ ,  $C_r^n$
- complement  $B - A$ ,  $B \setminus A$
- complete graph  $K_n$
- connected graph
- cycle
- degree (graph theory)  $\deg A$
- disjoint
- edge (graph theory)
- element
- empty set  $\emptyset$
- existential quantifier  $\exists$
- finite set
- infinite set
- intersection  $A \cap B$
- method of exhaustion
- order (set)  $|S|$
- path
- permutation  $P_r^n$
- proper subset  $\subsetneq$
- set  $\{\dots\}$
- simple graph
- subset  $\subset, \subseteq$



- symmetric difference  $A \Delta B$
- union  $A \cup B$
- universal quantifier  $\forall$
- vertex (graph theory)
- walk